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Date Functions

Important: Dates and times are converted to Coordinated Universal Time (UTC) in Archer database. As a result, dates and times in calculations are returned in UTC.

The following date functions produce dynamic values and to manipulate date information.

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DATEADD Function

The DATEADD function increases or decreases a date/time value by a given number of date/time units, such as days, hours or minutes.

Dates and times are converted to Coordinated Universal Time (UTC) in the Archer database. As a result, dates and times in calculations are returned in UTC.

Important: DATEADD always considers time, even if the referenced Date field is not configured to show time information. If a literal date string is supplied that does not contain time, midnight will be assumed.

Return Type: Date with time

Syntax: DATEADD(**datetime_unit**, **increment**, **datetime**)

In the above syntax, parameters in bold are required.

The following table describes DATEADD function parameters.

Parameter	Description
datetime_unit	The date/time part that should be used as the interval for increasing or decreasing the datetime parameter’s value. This parameter can be entered as DAY, HOUR or MINUTE.
increment	The number of date/time units that should be added to the datetime parameter’s value. This parameter must be formatted as a positive or negative integer greater than or equal to 1. (Decimal places are not supported.) If a positive number is provided, the function adds the specified number of date/time units to the datetime parameter’s value. If a negative number is provided, the function performs a subtraction.
datetime	The date/time value that should be increased or decreased by the specified number of date/time units. This parameter should be formatted as a Date-field reference, for example, [field name].

Examples:

The following table provides example formulas of the DATEADD function.

Formula	Result
DATEADD(DAY, 10, [First Published]) where the value of First Published is 8/10/2010 7:21 AM	8/20/2010 7:21 AM
DATEADD(HOUR, 6, [First Published]) where the value of First Published is 8/10/2010 7:21 AM	8/10/2010 1:21 PM

Formula	Result
DATEADD(MINUTE, 30, [First Published]) where the value of First Published is 8/10/2010 7:21 AM	8/10/2010 7:51 AM

DATEDIF Function

The DATEDIF function calculates the number of days between 2 dates.

Dates and times are converted to Coordinated Universal Time (UTC) in the Archer database. As a result, dates and times in calculations are returned in UTC.

Important: DATEDIF always considers time in the comparison, even if the referenced Date field is not configured to show time information. If a literal date string is supplied that does not contain time, midnight will be assumed.

Return Type: Numeric

Syntax: DATEDIF(**start_date**, **end_date**, datetime_unit)

In the above syntax, parameters in bold are required.

The following table describes DATEDIF function parameters.

Parameter	Description
start_date	The starting date of the period. This date can be entered as a hard-coded value, for example, 10/21/2010, or as a Date-field reference, for example, [date field name]. If a hard-coded value is supplied, it must be wrapped in the DATETIMEVALUE function. If time is supplied to DATETIMEVALUE in a date string, it must be in 24-hour clock format, for example, 14:25 represents 2:45 PM.
end_date	The ending date of the period. This date can be entered as a hard-coded value (for example, 10/21/2004) or as a Date-field reference (for example, [date field name]). If a hard-coded value is supplied, it must be wrapped in the DATETIMEVALUE function. If time is supplied to DATETIMEVALUE in a date string, it must be in 24-hour clock format, for example, 14:25 represents 2:45 PM.
datetime_unit	The granularity of the time information to be returned. This parameter can be entered as DAY, HOUR or MINUTE. If the datetime_unit parameter is omitted, DAY will be assumed. If DAY is specified, the difference will be calculated based on 24 hour periods, rather than the day portion of the date value.

Examples:

The following table provides example formulas of the DATEDIF function.

Formula	Result
DATEDIF(DATETIMEVALUE("10/21/2010"), [First Published]) where the value in the First Published field for the record is 11/26/2010.	36

Formula	Result
DATEDIF([First Published], [Last Updated], DAY) where the value in the First Published field for the record is 11/26/2010 11:59 PM and the value in the Last Updated field is 11/27/2010 12:01 AM. In this example, the day difference is zero (0) because the 2 dates are not 24 hours apart.	0
DATEDIF([First Published], [Last Updated], HOUR) where the value in the First Published field for the record is 10/1/2010 8:05 AM and the value in the Last Updated field is 10/3/2010 10:32 AM.	50
DATEDIF([First Published], [Last Updated], MINUTE) where the value in the First Published field for the record is 10/1/2010 8:05 AM and the value in the Last Updated field is 10/1/2010 10:32 AM.	147

DATEFORMAT Function

The DATEFORMAT function returns the supplied date in the format specified by the date mask.

Dates and times are converted to Coordinated Universal Time (UTC) in the Archer database. As a result, dates and times in calculations are returned in UTC.

Important: DATEFORMAT always permits full date and time formatting for the given Date field, even if that field is not configured to display time information.

Return Type: Text

Syntax: DATEFORMAT(**date**, **date_mask**)

In the above syntax, parameters in bold are required.

The following table describes DATEFORMAT function parameters.

Parameter	Description
date	The starting date of the period. This date can be entered as a hard-coded value, for example, 10/21/2010) or as a Date-field reference, for example, [field name].
date_mask	The mask used for formatting the returned date. The date_mask parameter must be enclosed in quotes. Date masks used with the DATEFORMAT function can contain any combination of the date elements.

The following table provides examples of possible date part combinations.

Date Mask	Return Example
M-d-yy h:mm tt	8-2-10 9:30 AM
MM.dd.yyyy	08.02.2010
MMMM d, yyyy HH:mm	August 2, 2010 09:30
yyyy-MM-dd	2010-08-02

Date Mask	Return Example
MMddyy	080210

The following separator characters are supported for date masks:

- space
- forward slash (/)
- hyphen (-)
- period (.)
- comma (,)
- colon (:

Examples:

The following table provides example formulas of the DATEFORMAT function.

Formula	Result
DATEFORMAT([First Published], "M/d/yyyy h:mm tt") where the date-time value of First Published is 8/2010 7:21 AM.	8/20/2010 7:21 AM
DATEFORMAT([Last Updated], "M/d/yyyy HH:mm") where the date-time value of Last Updated is 12/19/2010 2:51 PM.	12/19/2010 14:51
DATEFORMAT(NOW(), "h:mm tt") where the date-time value of NOW() is 8/6/2010 5:12 AM.	5:12 AM
DATEFORMAT([Start], "hh:mm t") where the date-time value of Start is 9/19/2010 6:48 PM.	06:48 P
DATEFORMAT([Stop], "H") where the date-time value of Stop is 4/8/2010 7:00 PM.	19
DATEFORMAT([Logged], "m") where the date-time value of Logged is 12/29/2010 3:57 PM.	57

DATETIMEVALUE Function

The DATETIMEVALUE function converts a literal date/time string to a serial number. The serial number represents the number of whole and partial days that have elapsed since January 1, 1900.

Important: Dates and times are converted to Coordinated Universal Time (UTC) in the Archer database. As a result, dates and times in calculations are returned in UTC.

Return Type: Number (serial number representing date and time)

Syntax: DATETIMEVALUE(datetime_string)

This function only accepts dates in the US format (MM/DD/YYYY). In the above syntax, parameters in bold are required.

The following table describes the DATETIMEVALUE function parameter.

Parameter	Description
datetime_string	The literal date/time string value to be converted. This cannot be a field reference.

Examples:

The following table provides example formulas of the DATETIMEVALUE function.

Formula	Result
DATETIMEVALUE("10/02/2010")	40453
DATETIMEVALUE("10/02/2010 01:50")	40453.08

DAY Function

The DAY function returns an integer between 1 and 31, which represents the day of the month for the specified date value.

Important: Dates and times are converted to Coordinated Universal Time (UTC) in the Archer database. As a result, dates and times in calculations are returned in UTC.

Return Type: Numeric

Syntax: DAY(**date**)

In the above syntax, parameters in bold are required.

The following table lists the DAY function parameter.

Parameter	Description
date	The date value to be evaluated in determining the day of the month. This parameter should be formatted as a Date-field reference, for example, [field name].

Example:

The following table provides an example formula of the DAY function.

Formula	Result
DAY([Logged]) where the value in the Logged field is 7/13/2010 10:45 AM.	13

HOURL Function

The HOURL function returns an integer between 0 and 23, which represents the hour of the day for the specified date value. Formula validation will fail for this function if the Time Information option is not enabled for the Date field referenced in the date parameter.

Return Type: Numeric

Syntax: HOURL(**date**)

In the above syntax, parameters in bold are required.

The following table describes the HOUR function parameter.

Parameter	Description
date	The date value to be evaluated in determining the hour of the day. This parameter should be formatted as a Date-field reference, for example, [field name].

Example:

The following table provides an example formula of the HOUR function.

Formula	Result
HOUR([Logged]) where the value in the Logged field is 7/13/2006 2:45 PM.	14

MINUTE Function

The MINUTE function returns an integer between 0 and 59, which represents the minute of the hour for the specified date value. Formula validation will fail for this function if the Time Information option is not enabled for the Date field referenced in the date parameter.

Important: Dates and times are converted to Coordinated Universal Time (UTC) in the Archer database. As a result, dates and times in calculations are returned in UTC.

Return Type: Numeric

Syntax: MINUTE(**date**)

In the above syntax, parameters in bold are required.

The following table describes the MINUTE function parameter.

Parameter	Description
date	The date value to be evaluated in determining the minute of the hour. This parameter should be formatted as a Date-field reference, for example, [field name].

Examples:

The following table provides example formulas of the MINUTE function.

Formula	Result
MINUTE([Logged]) where the value in the Logged field is 7/13/2006 2:45 PM.	45
MINUTE([Patch Date]) where the Patch Date field is a Date field that is not configured to accept time entry.	0

MONTH Function

The MONTH function returns an integer between 1 and 12, which represents the month of the year for the specified date value.

Important: Dates and times are converted to Coordinated Universal Time (UTC) in the Archer database. As a result, dates and times in calculations are returned in UTC.

Return Type: Numeric

Syntax: MONTH(**date**)

In the above syntax, parameters in bold are required.

The following table describes the MONTH function parameter.

Parameter	Description
date	The date value to be evaluated in determining the month of the year. This parameter should be formatted as a Date-field reference, for example, [field name].

Example:

The following table provides an example formula of the MONTH function.

Formula	Result
MONTH([Logged]) where the value in the Logged field is 7/13/2010 2:45 PM.	7

MONTHNAME Function

The MONTHNAME function returns the name of the month for the supplied date value. The return value is the full name, not an abbreviation.

Important: Dates and times are converted to Coordinated Universal Time (UTC) in the Archer database. As a result, dates and times in calculations are returned in UTC.

Return Type: Numeric

Syntax: MONTHNAME(**date**)

In the above syntax, parameters in bold are required.

The following table describes the MONTHNAME function parameter.

Parameter	Description
date	The date value to be evaluated in determining the month of the year. This parameter should be formatted as a Date-field reference, for example, [field name].

Example:

The following table provides an example formula of the MONTHNAME function.

Formula	Result
MONTHNAME([Due Date]) where the value in the Due Date field is 7/13/2010 2:45 PM.	July

NOW Function

The NOW function returns the current date/time. Each time a record is recalculated, the calculated field displays an updated date/time value. The full timestamp is stored for the calculated Date field even if the field is not configured to display time. If the Time Information option is later enabled for the field, the time will be displayed as it was originally computed.

Internally, the NOW function returns a serial number that represents the number of whole and partial days that have elapsed since January 1, 1900. From the user perspective, the value returned by the NOW function displays differently depending on the type of field to which the value will be returned.

Important: Dates and times are converted to Coordinated Universal Time (UTC) in the Archer database. As a result, dates and times in calculations are returned in UTC.

Return Type: Varies based upon the type of field receiving the return value. See the examples below.

Syntax: NOW()

This function does not have any parameters.

Examples:

The following table provides example formulas of the NOW function. For these examples, assume that the current date and time is October 2, 2010 at 1:46 a.m.

Field Type	Formula	Result
Numeric	NOW()	40453.073611111
Date	NOW()	10/02/2010 1:46 AM
Text	DATEFORMAT(NOW(),"M/d/yyyy h:mm tt")	10/02/2010 1:46 AM

QUARTER Function

The QUARTER function returns an integer between 1 and 4, which represents the calendar quarter in which the specified date value falls.

Important: Dates and times are converted to Coordinated Universal Time (UTC) in the Archer database. As a result, dates and times in calculations are returned in UTC.

Return Type: Numeric

Syntax: QUARTER(**date**)

In the above syntax, parameters in bold are required.

The following table describes the QUARTER function parameter.

Parameter	Description
date	The date value to be evaluated in determining the calendar quarter. This parameter should be formatted as a Date-field reference, for example, [field name].

Example:

The following table provides an example formula of the QUARTER function.

Formula	Result
QUARTER([Due Date]) where the value in the Due Date field is 12/15/2010 8:00 PM.	4

SYSTEMTODAY Function

The SYSTEMTODAY function returns the date value for the current date in the timezone configured for the particular instance in the Archer Control Panel. Each time the field is recalculated, the calculated field displays an updated date.

Internally, the SYSTEMTODAY function returns a serial number that represents the number of whole days that have elapsed since January 1, 1900.

The TODAY function returns the current day only in Coordinated Universal Time (UTC). The time component is not relevant and is set to 00:00. The SYSTEMTODAY function returns the current day only based on the instance timezone set in the Archer Control Panel.

Return Type: Varies based upon the type of field receiving the return value. See the examples below.

Syntax: SYSTEMTODAY()

This function does not have any parameters.

Examples:

The following table provides example formulas of the TODAY function. For these examples, assume that the current date and time is October 2, 2010 at 1:46 a.m.

Field Type	Formula	Result
Numeric	SYSTEMTODAY()	40453
Date	SYSTEMTODAY()	10/02/2010
Text	DATEFORMAT(SYSTEMTODAY(),"M/d/yyyy")	10/02/2010

TODAY Function

The TODAY function returns the date value for the current date. Each time a record is recalculated, the calculated field will display an updated date.

Internally, the TODAY function returns a serial number that represents the number of whole days that have elapsed since January 1, 1900. From the user perspective, the value returned by the TODAY function will display differently depending on the type of field to which the value will be returned.

Important: Dates and times are converted to Coordinated Universal Time (UTC) in the Archer database. As a result, dates and times in calculations are returned in UTC.

Return Type: Varies based upon the type of field receiving the return value. See the examples below.

Syntax: TODAY()

This function does not have any parameters.

Examples:

The following table provides example formulas of the TODAY function. For these examples, assume that the current date

and time is October 2, 2010 at 1:46 a.m.

Field Type	Formula	Result
Numeric	TODAY()	40453
Date	TODAY()	10/02/2010
Text	DATEFORMAT(TODAY(),"M/d/yyyy")	10/02/2010

WEEKDAY Function

The WEEKDAY function returns the day of the week for the supplied date value. The return value is the full name, not an abbreviation.

Important: Dates and times are converted to Coordinated Universal Time (UTC) in the Archer database. As a result, dates and times in calculations are returned in UTC.

Return Type: Text

Syntax: WEEKDAY(**date**)

In the above syntax, parameters in bold are required.

The following table describes the WEEKDAY function parameter.

Parameter	Description
date	The date value to be evaluated in determining the day of the week. This parameter should be formatted as a Date-field reference, for example, [field name].

Example:

The following table provides an example formula of the WEEKDAY function.

Formula	Result
WEEKDAY([Due Date]) where the value in the Due Date field is 12/15/2010 8:00 p.m.	Wednesday

WEEKNUMBER Function

The WEEKNUMBER function returns a number that indicates the week in which a given date falls for a calendar year beginning on January 1.

Important: Dates and times are converted to Coordinated Universal Time (UTC) in the Archer database. As a result, dates and times in calculations are returned in UTC.

Return Type: Numeric

Syntax: WEEKNUMBER(**date**, week_start)

In the above syntax, parameters in bold are required.

The following table describes WEEKNUMBER function parameters.

Parameter	Description

Parameter	Description
date	The date value to be evaluated in determining the day of the week. This parameter should be formatted as a Date-field reference, for example, [field name].
week_start	Accepts the keyword SUNDAY or MONDAY to specify whether weeks should be treated as beginning on Sunday or on Monday. If no value is passed for this parameter, SUNDAY will be assumed.

Examples:

The following table provides example formulas of WEEKNUMBER function.

Formula	Result
WEEKNUMBER([Due Date]) where the value in the Due Date field is 9/14/2008 (a Sunday).	38
WEEKNUMBER ([Due Date], SUNDAY) where the value in the Due Date field is 9/14/2008 (a Sunday).	38
WEEKNUMBER ([Due Date], MONDAY) where the value in the Due Date field is 9/14/2008 (a Sunday).	37

YEAR Function

The YEAR function returns the year corresponding to a date.

Important: Dates and times are converted to Coordinated Universal Time (UTC) in the Archer database. As a result, dates and times in calculations are returned in UTC.

Return Type: Numeric

Syntax: YEAR(**date**)

In the above syntax, parameters in bold are required.

The following table describes the YEAR function parameter.

Parameter	Description
date	The date of the year that you want to find. This parameter can be entered as a hard-coded date value, for example, "1/2/2010" or as a date-field reference, for example, [date field name].

Example:

The following table provides an example formula of the YEAR function.

Formula	Result
YEAR([First Published]) where the value in the First Published field is 11/26/2010.	2010

Date format descriptions

The following table describes date elements.

Date Element	Return Example
M	Displays the month as a number without a leading zero (Example: 1)
MM	Displays the month as a number with a leading zero (Example: 01)
MMMM	Displays the month as a full month name (Example: January)
d	Displays the day as a number without a leading zero (Example: 5)
dd	Displays the day as a number with a leading zero (Example: 05)
dddd	Displays the day as a full name (Example: Monday)
yy	Displays the year as a 2-digit number (Example: 06)
yyyy	Displays the year as a 4-digit number (Example: 2006)
h	Displays the hour as a 1-digit or 2-digit number based on a 12-hour clock format (Example: 9)
hh	Displays the hour as a 2 digit number (with a leading a leading zero, if necessary) based on a 12-hour clock format (Example: 09)
H	Displays the hour as a 1-digit or 2-digit number based on a 24-hour clock format (Example: 13)
HH	Displays the hour as a 2-digit number based on a 24-hour clock format (Example: 13)
m	Displays the minute as a number without leading zeros (Example: 5)
mm	Displays the minute as a number with leading zeros (Example: 05)
t	Displays the 1-letter AM/PM designator appropriate for the given time, regardless of whether the time is based on a 12-hour or 24-hour clock. (Example: 1:00 P for 12-hour clock; 13:00 P for 24-hour clock)
tt	Displays the 2-letter AM/PM designator appropriate for the given time, regardless of whether the time is based on a 12-hour or 24-hour clock. (Example: 1:00 PM for 12-hour clock; 13:00 PM for 24-hour clock)

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